Smart Irrigation Decision Agent

Project Description and Problem Statement

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Project description

Naba is a browser simulation for irrigation managers. It shows how six one-hectare farms can share one water supply over 30 days. Each farm has a limit on how much water its soil can hold, a target moisture level, and a lower level where crops begin to suffer.

The agent uses three AI pillars. It uses **knowledge** in the form of rules that check readings. It handles **uncertainty** by estimating moisture when a reading fails. It uses **optimization** to give the driest farms priority within the daily water limit.

The browser runs the rules and calculations without an AI API or a large language model (LLM). It shows how much water each farm gets and why.

Research problem

Soil moisture changes from day to day. A fixed schedule can leave one farm too dry and give another more water than it needs. Sensors help track those changes. A sensor can also get stuck on one value, report an impossible value, or stop sending readings.

When farms share a daily limit, giving one farm more water leaves less for the others. The agent must decide which readings to trust and which farms need water first.

Research question. How can a rule-based AI agent share a limited daily water supply across six farms when some soil moisture readings are wrong or missing?